Revision-Note Review Worksheet for Expert Rater

This booklet contains one student revision note to review. It is a take-home revision note prepared by a teacher for a JSS1 Mathematics class on Construction: parallel and perpendicular lines, bisection, and angles 90 and 60 degrees. Judge how well it covers the intended objectives and curriculum, follows the teacher's lesson plan, stays on topic and is conceptually correct, and is clear and useful for the students.

Lesson and revision-note context

Subject	Mathematics
Grade	JSS1
Term	Third Term
Week	6
Topic	Construction
Subtopic	Construction: parallel and perpendicular lines, bisection, and angles 90 and 60 degrees
Theme	—
Curriculum objectives	• construct parallel lines • construct perpendicular lines • bisect a given line segment • construct angle 90° • construct angle 60°
Previous knowledge	• Students have been taught how to draw lines.
Lesson sequence	• construct parallel lines — the teacher draws the construction of parallel lines on the board step by step, using a set square and ruler and explaining aloud as she draws; the student construct parallel lines, following the teacher's board construction and replicating it in their notes • construct perpendicular lines — the teacher draws perpendicular lines on the board step by step, using a set square and ruler and explaining aloud as she draws; the student follow the teacher's board construction and replicate it in their notes • bisect a given line segment — the teacher draws the bisection of a line segment on the board step by step with a pair of compasses, explaining aloud as she draws; the student construct line segment, following the teacher's board construction and replicating it in their notes • construct angle 90° — the teacher constructs angle 90 degrees on the board step by step by compass arcs from a point on the line, explaining aloud as she constructs; the student construct angle 90 degrees, following the teacher's board construction and replicating it in their notes • construct angle 60° — the teacher constructs angle 60 degrees on the board step by step by compass arcs of equal radius from the endpoint, explaining aloud as she constructs; the student construct angle 60 degrees, following the teacher's board construction and replicating it in their notes
Instructional materials	• Mathematical set.
Evaluation	—

Lesson objectives

1.	construct parallel lines
2.	construct perpendicular lines
3.	bisect a given line segment
4.	construct angle 90°
5.	construct angle 60°

Read the student revision note on the following pages, then complete the rating sheet at the end of this booklet.

STUDENT REVISION NOTE

Topic: Construction: parallel and perpendicular lines, bisection, and angles 90 and 60 degrees

Class: JSS1 **Subject:** Mathematics **Week:** 6

Section 1: Drawing Parallel Lines with a Set Square and Ruler

Two lines are parallel when they run side by side and never meet, no matter how far they are extended. This section shows you how to draw a line through a given point that stays parallel to a given line, using only your set square and ruler.

Here is the method for the problem: construct a line through point P parallel to the line AB.

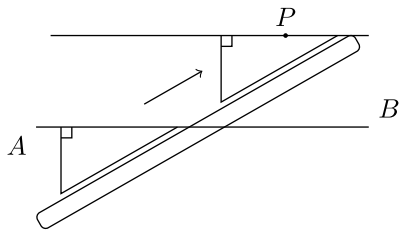
Step 1. Place the set square so that one of its edges lies exactly along AB (fig-01).

Step 2. Place the ruler firmly against one of the other edges of the set square. If you are right-handed, rest the ruler against the left-hand edge (fig-02). The ruler just holds the set square in place — its own slope does not matter, so the ruler may lean however is comfortable for you (fig-03). What matters is that the set square does not turn.

Step 3. Hold the ruler firmly and slide the set square along the ruler towards P. Keep sliding until the edge that was lying on AB reaches P (fig-04, fig-05).

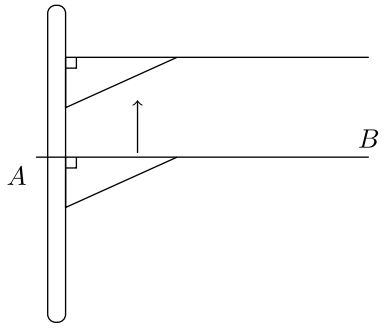
Step 4. Draw a line along that edge of the set square, through P. Because the set square only slid without turning, that edge keeps the same direction as it had on AB — so the new line is parallel to AB.

Visual



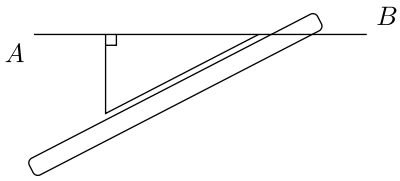
The same slide as fig-04, made against the leaning ruler of fig-03. The set square is drawn twice: once with its edge on AB, and once slid along the ruler until that edge passes through P. The two positions are congruent and point the same way — sliding along a held ruler moves the set square without turning it — so the edge through P stays parallel to AB. The arrow gives the direction of the slide, along the ruler's edge, and the right angle is marked in both positions.

Visual



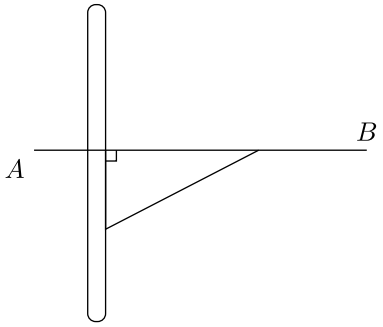
The vertical ruler held fixed, with the set square drawn twice — once with its edge on AB and once higher up, its edge on a second line parallel to AB — and an arrow between the two positions pointing up along the ruler.

Visual



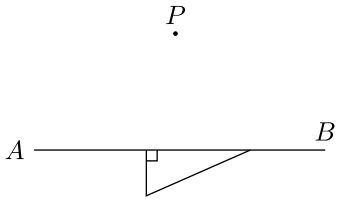
The same set square on AB as in fig-02, with the ruler moved to rest against its sloping edge instead of its upright one. The set square has not moved; only the ruler has, so it now leans across AB, running from lower left to upper right. The right angle is still where the edge on AB meets the short upright edge, and it is marked there. The ruler's inclination does not affect the construction. The set square is drawn at one particular size, and with its right angle at one particular end; neither is part of the method. Any set square works, either way round, so long as one edge lies along AB and the ruler rests against another.

Visual



The same set square on AB, with the ruler now drawn as a long strip lying along the set square's downward edge. The right angle marked at the corner is the set square's own, between the edge lying along AB and the edge running down from it.

Visual



Line AB, with a point P marked above it. A set square is drawn below AB with one edge along the line: the right angle is at the left end of that edge, one leg runs down from it, and the hypotenuse closes back to a point on AB to the right.

Worked example. Construct a line through point P that is parallel to the line AB.

Working through it:

1. Lay one edge of your set square exactly along AB. The edge must touch the whole line, and the right-angle corner sits at one end of that edge (fig-01).
2. Hold your ruler against the set square so it cannot slip — against the upright (left-hand) edge if you are right-handed (fig-02). Keep the ruler still from here on.
3. Slide the set square along the ruler, up towards P, without letting the set square turn. Stop as soon as the edge that was on AB passes through P (fig-05).
4. Draw a line along that edge through P. This line is parallel to AB.

Check your work: the two lines should look exactly side by side — the new line through P never leans towards or away from AB.

Practice

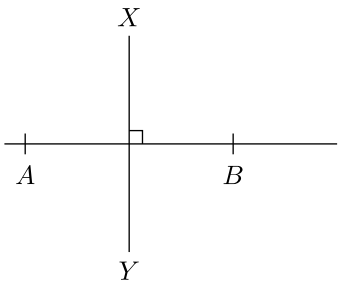
1. In the construction of a line through P parallel to AB, what must you do in Step 2 before you slide the set square, and why?
2. While sliding the set square along the ruler towards P, what tells you to stop, and what do you then draw?

3. Construct a line through point P that is parallel to the line CD, where C and D are the two end points of your line and P is a point marked above it. Use your set square and ruler, and describe the three moves you make.

Section 2: Constructing Perpendicular Lines

Two lines are said to be **perpendicular** to each other when they meet at a right angle — that is, an angle of exactly **90°**. Look at fig-06: line XY crosses line AB, and the small square mark drawn at the crossing shows the right angle between them. Because the angle at the crossing is a right angle, the two lines are perpendicular. Whenever you see that little square mark at a crossing, it is telling you the angle is $90\,^{\circ}$.

Visual



Line AB with a second line XY crossing it, X above and Y below, and a right-angle mark in the upper right quadrant at the crossing.

Construct a line perpendicular to line AB at point Q, using a set square and a ruler.

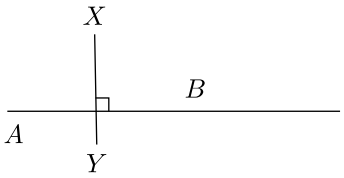
Step 1. Draw a line and label its ends A and B (fig-07). Mark a point Q somewhere on the line. This is the point where your new line must cross AB at a right angle.

Step 2. Place the edge of the set square along line AB, and let the edge of the ruler rest against the hypotenuse of the set square (fig-08). The ruler is now held still — it will guide the slide in the next step.

Step 3. Pressing firmly on the ruler, slide the set square down along the hypotenuse side until the third edge of the set square touches point Q. Keeping it there, draw a line XY along that edge (fig-09).

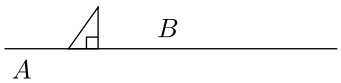
Line XY now crosses AB at Q with a right angle marked between them, so XY is perpendicular to AB.

Visual



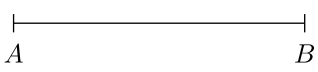
Line AB with XY now drawn across it, X above and Y below, and a right-angle mark in the upper right quadrant.

Visual



Line AB with the set square standing on it as a narrow right-angled triangle: a short base along AB, the upright edge rising from the right end of that base with the right angle marked there, and the hypotenuse closing back down to the left.

Visual



A plain line with a stroke at each end, labelled A on the left and B on the right.

Practice

1. Two lines meet at a crossing and the small square mark is drawn in the corner of that crossing. What does that square mark tell you about the angle between the two lines?
2. A line XY crosses line AB at point Q with a right angle at the crossing. What do we call lines XY and AB? Write down the steps you would follow with a set square and a ruler to draw XY.

Section 3: Bisecting a Line Segment

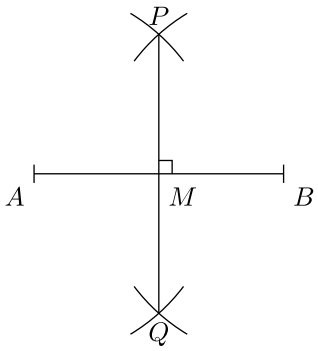
To **bisect** a line segment means to divide it into two equal halves. When a segment AB is bisected, the point where it is cut — call it M — is its **midpoint**, and the two parts AM and MB are exactly the same length, so $AM = MB$.

You can bisect a line segment using only a pair of compasses and a ruler. Here is how.

1. Draw the segment AB (fig-10).
2. Open your compasses so that the radius is about $\frac{3}{4}$ of the length of AB. Keep this radius the same for the whole construction.
3. Put the sharp point of the compasses on A and draw two arcs: one arc above AB and one arc below it.
4. Keeping the same radius, put the sharp point on B and draw two more arcs, one above and one below, so that they cut the first two arcs. The arcs crossing above meet at a point P, and the arcs crossing below meet at a point Q (fig-11).
5. Join P to Q with your ruler.
6. The line PQ crosses AB at M. Because the arcs were all drawn with the same radius, M is the midpoint of AB, so $AM = MB$. The line PQ also meets AB at a right angle ($90\,^\circ$), shown by the right-angle mark in fig-12.

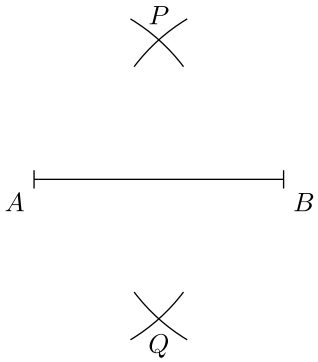
The secret of the whole construction is the single compass radius: drawing every arc with the same opening is what makes the crossing points P and Q sit exactly above and below the middle of AB.

Visual



The same arcs with P joined to Q, the joining line crossing AB at its middle at M and carrying a right-angle mark in the upper right quadrant.

Visual



The segment AB with arc pairs crossing above and below it, the upper crossing labelled P and the lower one Q. Each crossing is drawn as a small X where an arc from A meets an arc from B.

Visual



The line segment AB, a stroke at each end, A on the left and B on the right.

Worked example — bisecting a segment

Suppose you want to bisect a segment AB drawn on your page.

- Set your compasses to about $\frac{3}{4}$ of the length of AB and do not change this radius.
- Place the point on A. Draw one arc above AB and one arc below it.
- Place the point on B with the same radius. Draw one arc above and one below so they cut the arcs from A.
- The crossing above is P; the crossing below is Q (fig-11).
- Rule a straight line through P and Q.
- Where this line meets AB is M, the midpoint. Check it in fig-12: $AM = MB$, and PQ stands on AB at 90°.

Practice

1. What does it mean to bisect a line segment AB?
2. When bisecting a line segment AB, what should you keep the same about your compasses for all four arcs you draw?
3. Draw a segment AB. Bisect it using your compasses and ruler. Name the midpoint M, and state the two equal lengths you have created and the angle at which your bisecting line meets AB.

Section 4: Constructing a 90° Angle

A 90° angle is a right angle — the kind of corner you see at the junction of a horizontal line and a line rising straight up from it. You can build one using only your compasses and a ruler. The idea is to draw a perpendicular from a point on your line: a perpendicular is simply a line that meets another line at a right angle, so the angle it makes is exactly 90°.

You will need your mathematical set, and you should already know how to draw a straight line. Let's set out the construction.

Worked example — constructing angle 90° at a point on a line

We will construct angle 90° at a point on the line AB, following the figure (fig-13).

Step 1. Draw a straight line AB and mark a point C on it, where the 90° angle is to be constructed.

Step 2. With the compasses at C, draw an arc that cuts the line at two points. Call these points D and E.

Step 3. Open the compasses WIDER than the distance DC — the width used in step 2 will not work, because arcs drawn at that width meet only at C itself. With the compasses at D at this new wider setting, draw an arc above the line.

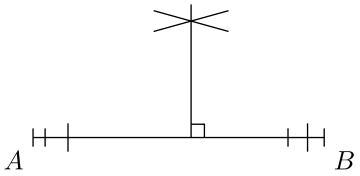
Step 4. Keeping the compasses at the wider setting of step 3, place them at E and draw another arc, cutting the first arc at F.

Step 5. Join C to F with your ruler.

Step 6. The line CF is perpendicular to AB, so $\angle FCB = 90\,^\circ$.

In fig-13 you can see the finished result: the perpendicular CF rises from a point near the middle of the line, the right-angle mark sits at its foot, and the two small strokes cross each other near the top of CF. The little square at the foot of the perpendicular is the right-angle mark — it shows the angle is 90°.

Visual



A line with a stroke at each end, A on the left and B on the right, and four short strokes standing on the line, a pair towards A and a pair towards B. A perpendicular rises from a point near the middle, with a right-angle mark at its foot, and two short strokes cross each other near its top.

Practice

1. Construct angle 90° at a point C on a straight line AB, using your compasses and ruler. In your construction, what is the special name given to the line you draw through C and F?
2. In the construction of angle 90°, after drawing the first arc from C that cuts the line at D and E, you must open the compasses wider than DC before drawing the next two arcs. Why is the step-2 width not enough for these arcs?
3. Draw a line AB, mark a point C on it, and construct the perpendicular to AB at C following the six steps. What is the measure, in degrees, of the angle between AB and the perpendicular you drew?

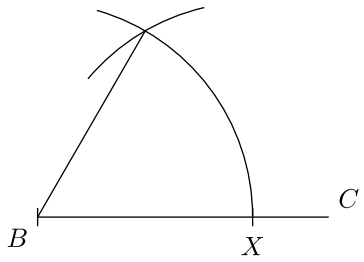
Section 5: Constructing an Angle of 60°

You already know how to draw straight lines. Now you will use a pair of compasses to construct an angle of exactly 60° — one of the most useful angles in construction, because it is the angle inside an equilateral triangle.

The key idea is that an equilateral triangle has all three sides equal, and all three angles are 60°. So if you can draw a triangle whose three sides are equal, you have drawn three 60° angles without measuring a single one. The compass does the work for you: wherever you draw two arcs with the same radius, you are marking off equal distances.

Here is how the construction works, using figure fig-14 as your guide. You start with a line segment BC and mark a point X on it. One compass radius is chosen once and used for every arc — that is what keeps the distances equal.

Visual



The line BC with B at the left, a stroke marking X part way along, a long arc swung from B through X and up to the left, a shorter arc swung from X cutting it, and a straight line from B to that crossing.

Worked example — construct an angle of 60° at point B.

Step 1. Draw a line segment BC. Open your compasses to any convenient radius. Place the point on B and swing an arc that cuts BC at a point you call X. In fig-14, X lies part way along the line, and the long arc from B swings up to the left through X.

Step 2. Keep the compasses at the same radius — do not change the opening. Place the point on X and swing a second arc. This arc cuts the first arc at a point A. (In fig-14 the shorter arc from X cuts the long arc from B at A.)

Step 3. Draw a straight line from B to A. The angle $\angle ABC$ (the angle at B between the line BA and the line BC) is exactly 60°.

Why is this correct? Because BX and XA are both drawn with the same compass radius, and BA joins B to A — so all three sides BX, XA and BA are equal. That makes triangle BXA equilateral, and every angle of an equilateral triangle is 60°. The angle you have constructed at B is $\angle XBA$, which is exactly that 60°.

Practice

1. Follow the three steps to construct an angle of 60° at B on a line segment BC. In your construction, why must the second arc (drawn with centre X) use exactly the same radius as the first arc (drawn with centre B)?
2. In the construction of a 60° angle, you first swing an arc from B that cuts the line at X, then swing a second arc from X. What point is marked where the second arc cuts the first arc, and what figure does joining B to that point complete?

Check your work

Attempt every question before you read this page.

Drawing Parallel Lines with a Set Square and Ruler

1. Place the ruler firmly against one of the set square's other edges (the left-hand edge if right-handed) and hold it still, so that when the set square slides along it, the set square does not turn.

   The ruler acts as a fixed track. If the ruler moved or the set square turned as it slid, the edge would no longer keep the direction it had along AB, and the final line through P would not be parallel to AB.
2. Stop when the edge of the set square that was lying on AB reaches P; then draw a line along that edge through P.

   The set square slides without turning, so the edge that matched AB keeps the same direction. When that edge passes through P, drawing along it gives the line through P parallel to AB.
3. The final line drawn through P is parallel to CD. The three moves are: (1) place one edge of the set square exactly along CD; (2) hold the ruler firmly against another edge of the set square; (3) slide the set square along the ruler until the edge reaches P, then draw the line through P along that edge.

Set the set square's edge on CD, brace the ruler against another edge and hold it still, then slide the set square up the ruler until the original edge passes through P. Drawing along that edge gives the line through P parallel to CD, because the slide kept the edge parallel to CD.

Constructing Perpendicular Lines

1. The square mark shows that the angle between the two lines is a right angle, exactly $90\,^\circ$, so the lines are perpendicular to each other.
2. XY and AB are perpendicular to each other.

   Steps: (1) Draw line AB and mark point Q on it. (2) Place the edge of the set square along line AB and rest the ruler against the hypotenuse of the set square. (3) Pressing firmly on the ruler, slide the set square down along the hypotenuse until its third edge touches point Q, then draw line XY along that edge.

Bisecting a Line Segment

1. It means to divide AB into two equal halves, so the point of bisection M is the midpoint and $AM = MB$.
2. The radius (the compass opening) must stay the same for every arc, set to about $\frac{3}{4}$ of the length of AB.
3. The bisecting line meets AB at M, so $AM = MB$, and it meets AB at $90\,^\circ$.

   Open the compasses to about $\frac{3}{4}$ of AB and keep that radius. From A draw an arc above and an arc below AB. With the same radius, from B draw an arc above and an arc below so they cut the first arcs; the upper crossing is P, the lower is Q. Rule a line through P and Q. Where it crosses AB is the midpoint M, giving $AM = MB$, and the two lines meet at $90\,^\circ$.

Constructing a 90° Angle

1. The line CF is a perpendicular to AB — that is why $\angle FCB = 90\,^\circ$.
2. Because arcs drawn at the step-2 width from D and E meet only at C itself, so they would not give a point F above the line to join to C.
3. $90\,^\circ$

Constructing an Angle of 60°

1. The second arc must use the same radius so that the distance XA equals the distance BX. Since BA joins B to A as well, all three sides BX, XA and BA become equal, making triangle BXA equilateral — and an equilateral triangle has all angles 60°. The angle you construct at B is therefore exactly 60°.

   Draw BC, place compass point on B and swing an arc to cut BC at X. Without changing the radius, place the point on X and swing a second arc to cut the first at A. Join BA. Because the radius was kept the same, BX = XA, and BA is the same straight distance again, so triangle BXA is equilateral and $\angle XBA = 60\,^\circ$.
2. The second arc cuts the first arc at A. Joining B to A completes triangle BXA, which is equilateral (all three sides equal), so the angle at B between BA and BC is 60°.

Rating sheet

Mark one score per criterion, then use the notes section for strengths, concerns, and recommendations. The lesson objectives are on the first page for reference.

Revision note ID	cand-b645b1d2	**Date**	
Subtopic	Construction: parallel and perpendicular lines, bisection, and angles 90 and 60 degrees	**Rater**	

> The reference sheet includes the teacher's evaluation questions. A pack that reuses them, reworded or verbatim, may be credited for matching how the lesson was assessed — but reuse on its own does not establish taught-lesson alignment. Rate alignment on the whole of it: the objectives covered, the order taught, what the activities had the students do, the method, and the expected depth.

Criterion (select one score per row)	**1** Poor	**2** Weak	**3** Adequate	**4** Good	**5** Excellent
Objective and curriculum alignment Does the revision note cover the intended lesson objectives and curriculum target?	○	○	○	○	○
Taught-lesson alignment Does it reflect the taught lesson — its sequence, emphasis, activities, and expected depth?	○	○	○	○	○
Scope and conceptual correctness Does it stay within the expected lesson boundary and avoid conceptual/mathematical errors?	○	○	○	○	○
Student-facing clarity and usefulness Can JSS1 learners understand and use it meaningfully as take-home review material?	○	○	○	○	○

Strengths	Concerns	Recommendation

Flag any problems (optional)

Tick all that apply. These are separate from the scores above — leave blank if none apply.

☐	**Misses a required objective** — Skips something the lesson was meant to teach.
☐	**Goes beyond this lesson** — Includes material outside this lesson's scope (e.g. next week's topic).
☐	**Wrong or misleading content** — Contains a mathematical or factual mistake.
☐	**Too hard for JSS1 to follow** — Likely confusing for the students.
☐	**Other:** _____